

Experiment No.3

Environment: Microsoft Windows

Tools/ Language: Oracle

OBJECTIVE: To implement the restrictions on the structure of the table.

Theory:

Data constraints: Besides the column name, column length and column data type, there are other parameters i.e. other data constraints that can be passed by the DBA at check creation time. The constraints can either be placed at column level or at the table level.

- i Column Level Constraints:** If the constraints are defined along with the column definition, it is called a column level constraint.
- ii Table Level Constraints:** If the data constraint attached to a specified column in a table reference the contents of another column in the table then the user will have to use table level constraints.

List of most used Constraint

- NOT NULL
- DEFAULT
- UNIQUE
- CHECK
- PRIMARY KEY
- FOREIGN KEY

Practical Assignment - 3

Department: Computer Engineering & Applications

Course: BCA

Subject: Database Management System Lab (BCAC-0816)

Year: 1ST

Semester: 2ND



1. Create the following tables and specify constraints at the time of creation.

Department

Column Name	Data Type	Size	Constraint
Deptno	number	3	primary key
Dname	varchar2	20	Unique
Location	varchar2	20	not null, department are located in Delhi, Pune, Agra

Employee

Column Name	Data Type	Size	Constraint
Empno	varchar2	5	primary key, should start with 'E'
Ename	varchar2	20	Unique
Designation	varchar2	20	not null
Salary	number	10	default 25000, must lie between 15000 and 50000
DOB	date		not null
Dno	number	3	foreign key (references department)

Candidate

Column Name	Data type	Size	Constraints
Candidate_ID	Number	6	Primary key of the table
Candidate_Name	Varchar2	20	Not Null
Candidate_Email	Varchar2	30	Unique, Must have '@' followed by '.' in between the email
Candidate_Dept	Number	2	Default 'HR'
Manager_ID	Varchar2	30	It can take only those values which are present in Candidate_ID column

2. Create the schemas as specified above without specifying any constraints.

College (cName : varchar2(10), c state : varchar2(10), enrollment : int) Student (sID : int, sName : varchar2(10), CGPA : number(2,1), marks :int) Apply (sID : int, cName : varchar2(10), major :
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- (i) Add **cName** as **Primary key** in **College**.
- (ii) Add **sID** as **Primarykey** in **Student**.
- (iii) Add **sID**, **cName**, **major** as **Primary key** in **Apply**.
- (iv) Add a new column **decision** in the **Apply** table keeping a constraint of **not null** for this column with data type **varchar2(3)**.
- (v) Change data type of **decision** in **Apply** to **char(1)**.
- (vi) Drop foreign key on column name **cName** from **Apply** table.
- (vii) Remove column **CGHS** from **Student** table.
- (viii) Drop **primary key** from **College**
- (ix) Modify foreign key on **sID** in **Apply** table to **foreign key on delete set null**.
- (x) Rename column **enrollment** to **enroll** in **College Table**.